

Modeling 2D Marshak Waves

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MOTIVATION

- Cylindrical radiation diffusion problems arise in many physical scenarios.
- The 1D solution only provides centerline front position $z_F(t)$.
- Real experiments display complicated 2D geometry requiring rigorous modeling.

THE MECHANISM

- Radiation propagates axially (along z) and radially (along r).
- Wall absorption causes overall energy loss.
- Radial temperature gradients dynamically develop.
- The radiation-driven heat front curves in (r, z) space.

PREVIOUS WORKS

- R. E. Marshak 1958 - First description and super-sonic solutions.
- J. H. Hammer; M. D. Rosen 2006 - New 1D solutions based on a perturbation theory.
- O. A. Hurricane; J. H. Hammer 2006 - Mathematical characterization of marshak waves curvature.
- A. P. Cohen; G. Malamud, S. I. Heizler 2020 - Modeling the 1D heat front Based on many experiments.

OUR WORK - OVERVIEW

- The idea: Solve the 1D propagation and radial perturbation structure separately and patch them.
- We deploy separation of variables on the 2D energy equation.
- The fundamental Bessel mode is introduced mathematically.
- The radial structure of the heat front elegantly surfaces as an analytical Bessel profile.

ASSUMPTIONS

- **Geometry:** Domain spans $0 \leq r \leq R$ (radius) and $0 \leq z \leq L$ (length).
- **Boundary Condition at Source:** Marshak boundary condition relates incoming and outgoing radiation safely at $z = 0$.
- **Wall Interaction:** Parameterized via Albedo (reflectivity factor α_{wall}).
- **Product Solution:** We adopt a separated solution profile:
$$T(r, z, t) = T_0(z, t) \cdot \Phi(r).$$

STATEMENT OF THE PROBLEM

The radiation diffusion equation in foam material:

$$\frac{\partial U}{\partial t} = \nabla \cdot (D \nabla U) \quad (1)$$

where $U = aT^4$ is the radiation energy density, and $D = \frac{c\lambda}{3}$ is the radiation diffusion coefficient.

In cylindrical coordinates (r, z) :

$$\frac{\partial U}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D \frac{\partial U}{\partial r} \right) + \frac{\partial}{\partial z} \left(D \frac{\partial U}{\partial z} \right) \quad (2)$$

THE 1D CENTERLINE SOLUTION

For the self-similar heat front at the exact centerline ($r = 0$):

$$x_F^2(t) = \frac{2 + \epsilon}{1 - \epsilon} \cdot C(\rho) \cdot H(t)^{-\epsilon} \cdot \int_0^t H(t') dt' \quad (3)$$

An implicit time-marching numerical approach acts on:

$$E(t_i) = E(t_{i-1}) + \frac{1}{2} [F_{\text{net}}(t_i) + F_{\text{net}}(t_{i-1})] \Delta t \quad (4)$$

incorporating wall energy loss $F_{\text{net}}(t_i) = F(0, t_i) - \frac{\dot{E}_{\text{wall}}(t_i)}{A}$.

WALL ENERGY LOSS & ALBEDO

Energy transfer to the cylindrical wall at $r = R$:

$$\frac{dE_{\text{wall}}}{dt} = 2\pi RL\sigma_{SB} \left(T_s^4 - \frac{\dot{E}_{\text{wall}}}{4\pi RL\sigma_{SB}} \right) \quad (5)$$

The albedo α_{wall} tracks the non-linear relationship between reflected and incident flux:

$$\alpha_{\text{wall}} = \frac{F_{\text{in}}}{F_{\text{out}}} = \frac{\sigma_{SB} T_s^4 + \frac{\dot{E}_{\text{wall}}}{2}}{\sigma_{SB} T_s^4 - \frac{\dot{E}_{\text{wall}}}{2}} \quad (6)$$

THE 2D RADIAL EIGENVALUE PROBLEM

Under assumed product formulation, the radial variant behaves linearly:

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{d\Phi}{dr} \right) + \kappa^2 \Phi = 0 \quad (7)$$

A Robin boundary condition dictates behavior at the wall $r = R$, securing a continuous profile without divergence. Thus $\Phi(r) = J_0(\kappa r)$.

The boundary enforces a transcendental equation for eigenvalues κ :

$$\kappa R \cdot J_1(\kappa R) = \epsilon_{\text{loss}} \cdot J_0(\kappa R) \quad (8)$$

RESULTS: CURVED FRONT POSITION

Two-Dimensional Temperature Distribution: The front curves precisely via the first eigenvalue κ_0 :

$$z_F(r, t) = z_F(t) \cdot J_0(\kappa_0 r) \quad (9)$$

Key observations:

- The front exhibits max penetration directly on axis ($r = 0$).
- **Perfect Reflector** ($\alpha_{\text{wall}} \rightarrow 1$): Reduces $\kappa_0 \rightarrow 0$ generating a decoupled flat 1D plane.
- **Total Absorber** ($\alpha_{\text{wall}} \rightarrow 0$): Compresses the function giving sharp decay off-axis.

RESULTS: REPRESENTATIVE SIMULATION

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update the path.

Figure: Self-Similar Solution of the 2D Heat Wave plotted against time.

COMPARISON TO NUMERICAL SIMULATIONS

(1)

- The analytically derived J_0 mapping maintains remarkably solid alignment with rigorous 2D simulations.
- Insert actual simulation plots here proving $T(r, z, t)$ validation.

COMPARISON TO NUMERICAL SIMULATIONS

(2)

- Highlight variations of albedo.
- Discuss errors mapping extreme ϵ shifts.

SUMMARY

- We offer an explicit analytical derivation coupling the 1D self-similar wave front equations onto 2D bounded spaces.
- Separation of variables generates a natural $J_0(\kappa_0 r)$ curvature.
- The model elegantly wraps radiation losses through empirical albedo adjustments across the wall.
- Accurate, cheap, and easily adapted for varying pulse profile shapes.